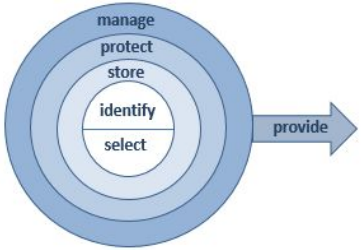
Big Data to Knowledge(BD2K)

Best Practices for Biomedical Research Data Management

Module 8

Curation & Preservation for Data



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Learning Objectives

1. Emphasize characteristics of long-term data curation and preservation that build on and extend active data management
2. Distinguish between preservation and curation
3. Explore the data curation role and its importance throughout the research data life cycle
4. Review good practices for data curation and preservation

Digital Content Lifecycle

Identify: what digital content do we have?

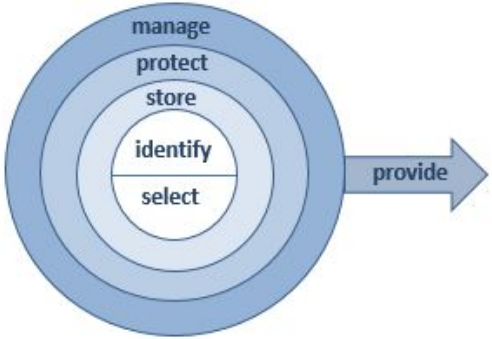
Select: what portion will be preserved?

Store: are there issues for long term storage?

Protect: what steps are needed to protect digital content?

Manage: are there provisions for long-term management?

Provide: are there considerations for long-term access?



Digital Preservation Outreach and Education: DPOE Baseline Digital Preservation Curriculum
http://www.digitalpreservation.gov/education/ttt.html

Curation/Preservation

Digital Stewardship: contributions to the longevity and usefulness of digital content by its caretakers that may occur within, but often outside of, a formal digital preservation program

Digital Curation: the combination of data curation and digital preservation

Data Curation: the value-added activities and features that stewards of digital content engage in to make digital content meaningful or useful

Digital Preservation: all of the activities an organization or individual engages in to ensure the longterm readability and usefulness of specified digital content

DP Standards and Practice

TDR: Trusted Digital Repositories, 2002

OAIS: Open Archival Information System Reference Model, ISO 14721: 2003 & 2012 & 2017 pending

PAIMAS: Producer Archive Interface Method Abstract Standard (PAIMAS), ISO 20652:2006; PAIS (specification), ISO 20104:2015

NISO Building Good Digital Collections, v3.0 2007

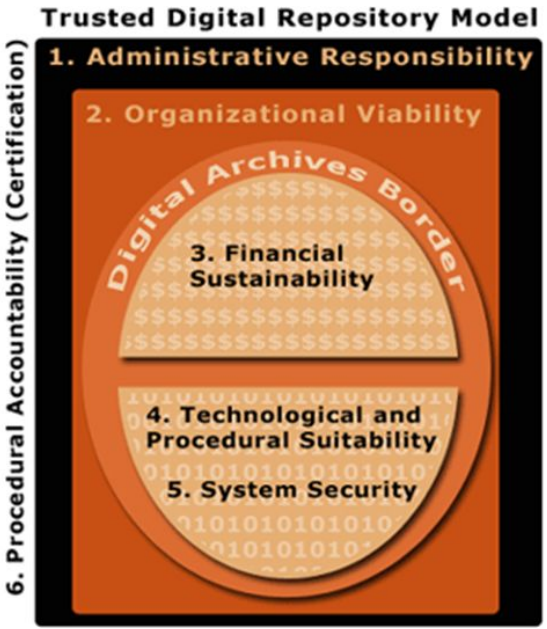
PREMIS: Preservation Metadata Implementation Strategies, 2005 plus updates (v.3.0 in 2015)

BRTF: Blue Ribbon Task Force on Sustainable Preservation and Access, 2010

TRAC: Trustworthy Repositories Audit and Certification, 2007 and ISO 16363: 2012 & revision pending …

Trusted Digital Repository

"A trusted digital repository is one whose mission is to provide reliable, long-term access to managed digital resources to its designated community, now and in the future."



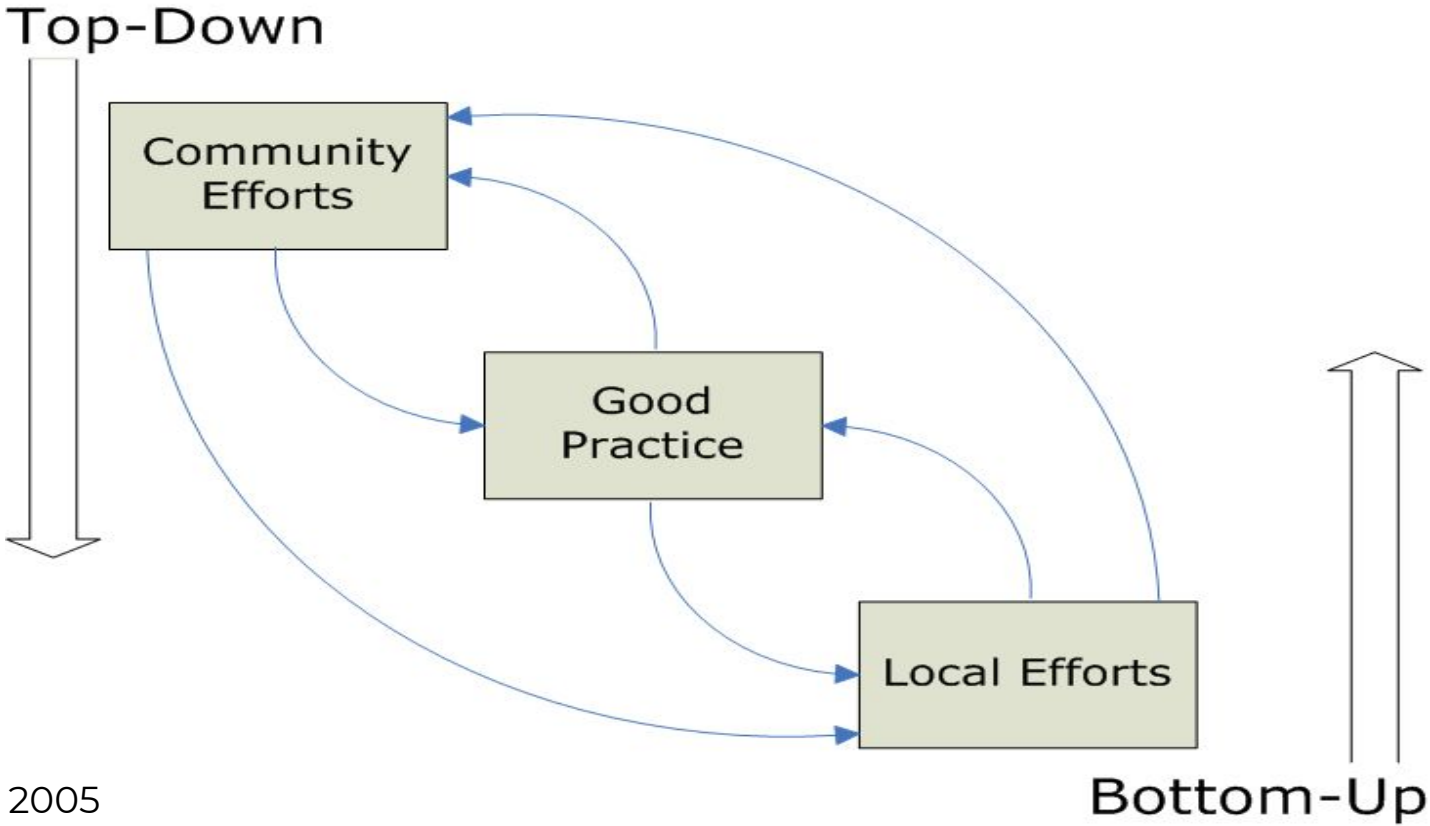
This Digital Preservation Management (DPM) model developed by Kenney and McGovern is grounded in the TDR attributes.

Research Libraries Group. (2002). *Trusted digital repositories: attributes and responsibilities an RLG-OCLC report*. Mountain View, CA: Research Libraries Group.
http://www.oclc.org/content/dam/research/activities/trustedrep/repositories.pdf

Open Archival Information System

Open Archival Information System (OAIS) Model

Emerging Standards & Practice



McGovern, 2005

Related Topics

- Institutional retention policies
- Data retention benefits and challenges
- Appraisal of research data
- Records within the research context
- Data and repositories